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National University of Computer and Emerging Sciences [redacted]

Lahore Campus [redacted]

Engineering Economics (MG3008)

Final Exam

Date: May 25th 2026

Course Instructor(s)

Mr. Syed Haris Mujtaba

Total Time (Hrs):	3
Total Marks:	120
Total Questions:	5

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Student Signature

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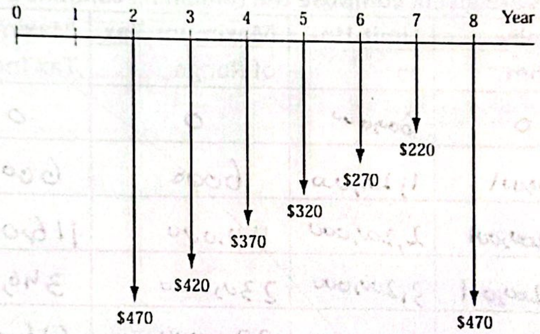
Attempt all the questions. Show each step and all the calculations.

In those question where the table is provided fill the values in them and return the question paper with answer sheet.

CLO #: 2 Interpret the concept of time value of money

Q1: By applying time value of money for the cash flows shown in the diagram given below, determine the present worth in year 0 at an interest rate of 10% per year. **[20 marks]**



CLO #: 3 Analyze cost effectiveness of alternatives using engineering economy methods

Q2: A consulting engineer is analyzing two methods for lining ponds used for evaporating concentrate generated during reverse osmosis treatment of brackinsh ground water for the clay county industrial park. A geosynthetic bentonite clay liner, good for 3 years, will require a boiler costing $920 plus valves costing $360. The cost of water, treatment charges, electricity, etc will be $3.1 per hour. Alternatively a high-density polyethylene geomembrane will cost $3850 to buy, will have a useful life of 5 years, and will cost $1.28 per hour to operate. What is the minimum number of hours per year that the cooling system must be used in order to justify purchase of the geosynthetic bentonite clay liner system? The MARR is 10% per year, and the salvage values are negligible. **[20 marks]**

CLO #: 3 Analyze cost effectiveness of alternatives using engineering economy methods

Q3: A construction company bought a 180,000 metric ton earth sifter at a cost of $65,000. The company expects to keep the equipment a maximum of 5 years. The operating cost is expected to follow the series described by 40,000 + 10,000k, where k is the number of years since it was purchased (k=1,2,3,4,5). The salvage value is estimated to be $30,000 for years1 and 2 and $20,000 for years3 through 5. At an interest rate of 10% per year, **Compute** the economic service life and the associated equivalent annual cost of the sifter. **[20 marks]**

CLO #: 4 Develop cash flow sequences that include the effects of taxes and depreciation

Q4-a: Construct a loan repayment table for an asset worth $1,500,000, interest is 14% per year for 6 years; when equal end of year payments are made. **[15marks]**

N	Big-Bal	Interest	Payment	Principal	Remaining	Acc- Principal
0					1,500,000	
1	1,500,000	210,000	385,736	175,736	1,324,264	175,736
2	1,324,264	185,397	388,736	200,339	1,123,925	376,075
3	1,123,925	157,350	385,736	228,386	895,539	604,461
4	895,539	125,375	385,736	260,361	635,178	864,822
5	635,178	88,925	385,736	296,811	338,367	1,161,633
6	338,367	47,371	385,736	338,365	2.38 ≈ 0	1,499,998

≈ 1.5 mill

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Q4-b: The Marginal tax rates are given, **compose** the remaining columns of the table: **[5 Marks]**

Range	Tax Rate	Limit-From	Limit-Up-to	Maximum Tax of Range	Maximum Tax Incurred
600,000	0%	0	600,000	0	0
600,000	1%	600,001	1,200,000	6000	6000
1,000,000	11%	1,200,001	2,200,000	110,000	116,000
1,000,000	23%	2,200,001	3,200,000	230,000	346,000
900,000	30%	3,200,001	4,100,000	270,000	616,000
Above	35%	4,100,001	Above	Above	Above.

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CLO #: 4 Develop cash flow sequences that include the effects of taxes and depreciation

Q5-a: Create a MACRS table with 200% DB also calculate the depreciation amount and book value for each year of an asset, bought for $1,500,000 and has a life of 5 years. Bal / N* MACRS X 1.5 mill **[20 marks]**

N	Bal	Dep1	Remain	N*	Straight	Dep	Currency	BV
1	100	20	80			20	300,000	1,200,000
2	80	32	48	4.5	17.78	32	480,000	720,000
3	48	19.2	28.8	3.5	13.71	19.2	288,000	432,000
4	28.8	11.52	17.28	2.5	11.52	11.52	172,800	259,200
5	17.28	6.912	10.36	1.5	11.52	11.52	172,800	86,400
6	10.368	10.368	0	0.5	5.76	5.76	86,400	0

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Q5-b: Carry Forward from Q4 and Q5(a): From information calculated in table for interest in Q4-a; also use the **MACRS** table of Q5-a, and make the rest of the income statement using the tax table in Q4-b, with Revenue, Material, Labor and Overhead and their relative inflations given below: **Assemble** the complete table of income statement. **[20marks]**

	"f"	0	1	2	3	4	5
Revenue	15	2,500,000	2,875,000	3,306,250	3,802,188	4,322,516	5,028,393
Material	9	200,000	218,000	237,620	259,066	282,382	307,796
Labor	8	300,000	324,000	349,920	377,914	408,147	440,799
Overhead	8	100,000	108,000	116,640	125,971	136,049	146,933
RMC			2,225,000	2,602,070	3,039,271	3,545,938	4,132,865
Dep			300,000	480,000	288,000	172,800	172,800
EBIT			1,925,000	2,122,070	2,751,297	3,373,138	3,960,068
Interest			210,000	185,397	157,350	125,375	88,925
EBT			1,715,000	1,936,673	2,593,947	3,247,763	3,871,140
Tax			62,650	87,034	206,608	360,329	547,342
EAT			1,652,350	1,849,639	2,387,339	2,887,434	3,323,798

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